

# Sysinternals Suite Lab Practical (CDAC Digital Forensics)

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## Experiment 1: Process Monitor (Procmon)

### Aim

To monitor real-time file system, registry, process, and thread activities.

### Tools Required

- Windows 10/11
- Sysinternals Suite
- Administrator privileges

### Procedure

1. Open PowerShell as Administrator.
2. Navigate to the Sysinternals folder:

```
cd "C:\Users\CDAC\Downloads\SysinternalsSuite"
```

3. Run:

```
.\Procmon64.exe
```

4. Accept the EULA.
5. Press **Ctrl + E** to stop capture.
6. Press **Ctrl + X** to clear events.
7. Press **Ctrl + E** to start capturing.
8. Open Notepad, create a file named **Test.txt**, type some text, save it, and close Notepad.

9. Stop capture (**Ctrl + E**).

10. Filter events:

- Filter → Process Name → is → `notepad.exe` → Include.

## Expected Output

- `CreateFile`
- `WriteFile`
- `ReadFile`
- `CloseFile`
- `QueryInformationFile`

## Observation

Process Monitor records all file operations performed by Notepad.

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# Experiment 2: Process Explorer

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## Aim

To analyze running processes and their resources.

## Procedure

1. Run:

```
.\procexp64.exe
```

2. Open Notepad.

3. Locate **notepad.exe** in the process list.

4. Double-click it.

## Observe

- Process ID (PID)

- Parent Process
- CPU Usage
- Memory Usage
- Threads
- Handles
- Loaded DLLs

## Additional Task

Enable:

Options → Verify Image Signatures

Observe which processes are digitally signed.

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# Experiment 3: Autoruns

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## Aim

To identify programs that start automatically.

## Procedure

1. Run:

```
.\Autoruns64.exe
```

2. Wait for the scan to complete.

3. Select:

Options → Hide Microsoft Entries

4. Refresh (F5).

## Observe

- Startup applications
- Services
- Drivers

- Scheduled Tasks
- Explorer extensions

## Task

Disable one non-essential startup item and re-enable it after observation.

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# Experiment 4: TCPView

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## Aim

To monitor active network connections.

## Procedure

1. Run:

```
.\tcpview64.exe
```

2. Open a web browser and visit any website.

3. Observe new TCP connections.

## Observe

- Process Name
  - Local Address
  - Local Port
  - Remote Address
  - Remote Port
  - Connection State
- 

# Experiment 5: Sigcheck

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## Aim

To verify digital signatures of executable files.

## Procedure

Run:

```
.\sigcheck64.exe C:\Windows\System32\notepad.exe
```

## Observe

- Publisher
  - Digital Signature
  - SHA256 Hash
  - File Version
- 

# Experiment 6: Strings

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## Aim

To extract printable strings from executable files.

## Procedure

```
.\strings64.exe C:\Windows\System32\notepad.exe
```

## Observe

- File names
  - Registry paths
  - URLs
  - DLL names
  - Readable text
- 

# Experiment 7: Handle

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## Aim

To identify which process is using a file.

## Procedure

1. Open `Test.txt` in Notepad.
2. Run:

```
.\handle64.exe Test.txt
```

## Observe

- Process Name
  - PID
  - File Handle
- 

# Experiment 8: PsList

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## Aim

To display all running processes.

## Procedure

```
.\pslist64.exe
```

## Observe

- Process Name
  - PID
  - CPU Time
  - Threads
  - Memory Usage
-

# Experiment 9: PsInfo

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## Aim

To collect system information.

## Procedure

```
.\PsInfo64.exe
```

## Observe

- Operating System
  - Processor
  - Installed RAM
  - Uptime
  - Hotfixes
- 

# Experiment 10: PsExec

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## Aim

To execute commands with SYSTEM privileges.

## Procedure

Display help:

```
.\PsExec64.exe
```

Open a SYSTEM command prompt:

```
.\PsExec64.exe -i -s cmd.exe
```

# Observe

A new Command Prompt opens as **NT AUTHORITY\SYSTEM**.

## Observation Table

| Experiment | Tool             | Observation                                   |
|------------|------------------|-----------------------------------------------|
| 1          | Process Monitor  | File, registry, process, and thread events    |
| 2          | Process Explorer | Running processes, CPU, memory, DLLs, handles |
| 3          | Autoruns         | Startup programs and persistence locations    |
| 4          | TCPView          | Active TCP/UDP connections                    |
| 5          | Sigcheck         | Digital signatures and file hashes            |
| 6          | Strings          | Readable strings from executables             |
| 7          | Handle           | Processes holding file handles                |
| 8          | PsList           | Running processes and statistics              |
| 9          | PsInfo           | System hardware and OS information            |
| 10         | PsExec           | Local/remote command execution                |

## Learning Outcomes

After completing these practicals, students will be able to:

- Monitor real-time system activity using **Process Monitor**.
- Analyze running processes with **Process Explorer**.
- Detect startup persistence using **Autoruns**.
- Monitor network connections using **TCPView**.
- Verify executable authenticity with **Sigcheck**.
- Extract readable strings from binaries using **Strings**.
- Identify processes locking files using **Handle**.
- List active processes with **PsList**.
- Collect detailed system information with **PsInfo**.
- Execute commands with elevated privileges using **PsExec**.



